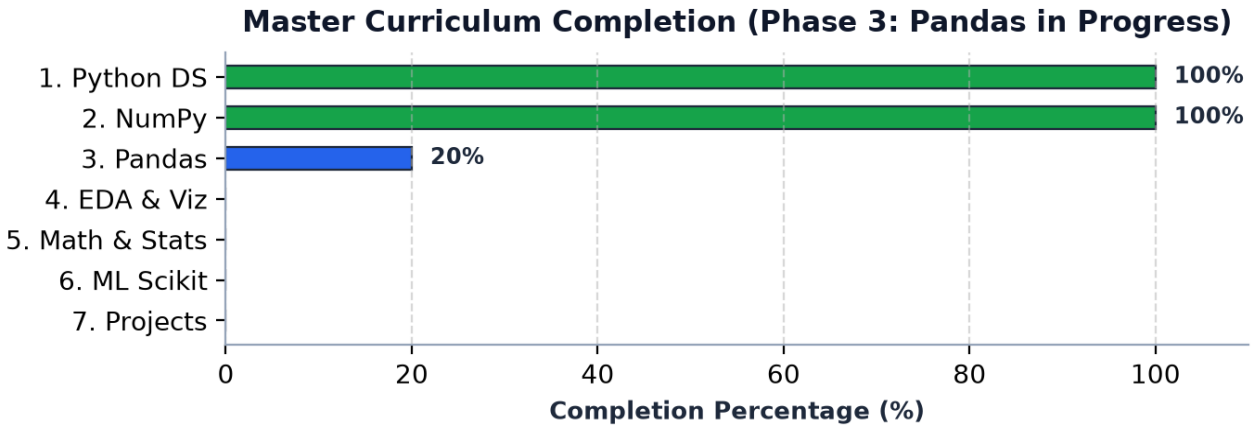
■ Data Science Long-Term Learning Progress Tracker

Student: **Amit** | Track: **Data Science & Practical Machine Learning** | Status: **Phase 3: Pandas In Progress (28/28 Solved! ■)**



■■ Curriculum Roadmap Status

Phase	Topic Domain	Problems Solved	Status
Phase 1	Python Core & Memory Architecture	12 / 12	**COMPLETED (100% Mastery) ■**
Phase 2	NumPy Vectorization & Dispersion	12 / 12	**COMPLETED (100% Mastery) ■**
Phase 3	Pandas Data Manipulation & Wrangling	4 / 20	**IN PROGRESS (Topic 3.1 Mastered: 20%) ■**
Phase 4	EDA & Visualization (Seaborn/Matplotlib)	0 / --	Pending Phase 3
Phase 5	Applied Statistics & Probability	0 / --	Pending Phase 4
Phase 6	Classical ML (Scikit-Learn)	0 / --	Pending Phase 5
Phase 7	End-to-End Capstone Pipelines	0 / --	Pending Phase 6

■ Practice Performance Log (28 / 28 Problems Mastered)

Set #	Topic Focus	Problems	Accuracy	Archive Location
Set 1.1	Memory References, Slicing, Safe Dicts	4 / 4	**100%**	practice_archive/Session_01/Set_01
Set 1.2	Pipelines, Lambdas, Sanitizers	4 / 4	**100%**	practice_archive/Session_01/Set_02
Set 1.3	File I/O, CSV DictReader, JSON load	4 / 4	**100%**	practice_archive/Session_01/Set_03
Set 2.1	NumPy Ndarray, SIMD, arange/linspace	4 / 4	**100%**	practice_archive/Session_02/Set_01
Set 2.2	2D Slicing, Axis=0/1, Boolean Masking	4 / 4	**100%**	practice_archive/Session_02/Set_02
Set 2.3	Z-Scores, Argmax, Reshape, Broadcasting	4 / 4	**100%**	practice_archive/Session_02/Set_03
Set 3.1	Series, DataFrames, Alignment, DTI Ratio	4 / 4	**100%**	practice_archive/Session_03/Set_01

■ Study Memory & Milestone Log

Session 1 (Python Core & Pipelines): Mastered RAM pointers vs deep copies, half-open interval indexing, and safe file I/O with context managers.
Session 2 (NumPy Foundations & 2D Slicing): Mastered contiguous C-memory layout, SIMD vectorization, 2D matrix sub-box extraction, and the axis collapse mental model (axis=0 vertical drop vs axis=1 horizontal walk).
Session 3 (NumPy Statistics & Broadcasting Engine): Mastered Z-score anomaly detection, np.argmax across axes, reshaping memory layout with -1 wildcard, and the NumPy broadcasting engine for feature mean-centering.
Session 4 (Pandas Architecture & Automatic Alignment): Unpacked the internal anatomy of pd.Series (NumPy C-buffer + Index hash-map) and

pd.DataFrame. Mastered label-based automatic alignment, .add(fill_value=0), .set_index(), and vectorized financial feature engineering.